

- (a) The alpine areas nearest to South Island are 1500–2000 km away across the sea. Many of South Island's alpine species live nowhere else in the world.

Explain how a large number of alpine plant species developed on South Island.

[5]

(b) Fig. 5.1 shows two aspects of the history of South Island over the last 3.9 million years.

- The dashed line shows how the mean height of mountains in the Clyde region of South Island increased over time. The mountains in this range have a mean height of 2400 m at the present time.
- The solid line models the height of the treeline over time based on geological climate data. The treeline was higher when the climate was warmer, and the treeline was lower when the climate was colder, during ice ages.

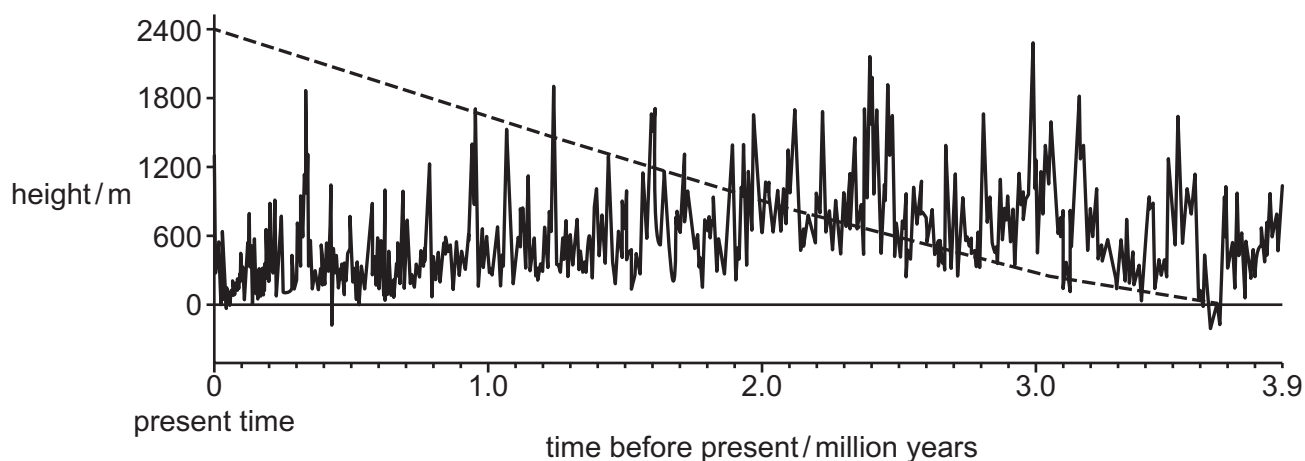


Fig. 5.1

- (i) With reference to Fig. 5.1, identify with reasons the time period when South Island's alpine plant species developed.

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- (ii) Suggest how DNA sequence data could be used to confirm the time period you identified in (i).

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[Total: 10]